Based on your Final Year Project (FYP) report, here is a structured 12-slide presentation outline in English. Since you have completed about **30% of the work** (UI development and backend preprocessing), the focus is shifted toward the technical implementation details and testing of these specific areas.

Slide 1: Title Slide

Project Title: Investigating Volumetric Analysis of Constituents of Brain Soft Tissue.

Team Members: Daniyal Ahmed Gul, Muhammad Hasan Nauman, Hadia Khan.

Supervisor: Dr. Masroor Ahmed.

Slide 2: Introduction - Problem & Significance

The Problem: Manual segmentation of brain tissues (Gray Matter, White Matter, CSF) is time-consuming, prone to human error, and computationally expensive.

Significance: Volumetric changes are key indicators for Alzheimer's, Multiple Sclerosis, and brain tumors.

Project Goal: To develop an automated, data-driven framework using 2D MRI slices to estimate 3D tissue volumes with high precision.

Slide 3: Introduction - Proposed Methodology

* **The Pipeline:**

Data Acquisition: Using public repositories like Kaggle, TCIA, and OASIS.

Preprocessing: Cleaning data via MATLAB/Python (resizing, denoising, skull stripping).

Core Engine: Hybrid feature extraction (Texture, Shape, and CNN features) followed by PCA for reduction.

Final Output: Volumetric reports and tissue classification using K-Means and ML models.

Slide 4: - Development Environment & Stack

OS: Windows-based environment.

IDE: Visual Studio Code.

Frontend: React and Next.js for a modern, component-based UI.

Backend: Python for heavy-duty MRI image processing (resizing, normalization, etc.).

Database: MongoDB with Mongoose for user/patient records.

Slide 5: [HEADING: USER INTERFACE - LANDING PAGE]

* *Add Screenshot of the Landing Page here.*

* **Key Features:**

* Provides entry points for researchers and new users.

* Highlights platform capabilities: Preprocessing, Volumetrics, and Classification.

Slide 6: [HEADING: USER INTERFACE - DASHBOARD & AUTH]

* *Add Screenshot of the Login/Signup and Dashboard Overview here.*

* **Functionality:**

* Secure authentication using **Bcrypt** password hashing.

* Dashboard provides "Quick Actions" for uploading images and generating reports.

* Summary of total analyses and success rates.

Slide 7: [HEADING: USER INTERFACE - MRI UPLOAD MODULE]

* *Add Screenshot of the Patient Info and File Upload screen here.*

* **Functionality:**

* Link MRI images to specific patient records.

* Supports multiple formats: JPG, PNG, DICOM, and NIfTI.

* Backend processes images and saves them to a temporary server directory.

Slide 8: Chapter 4 - Backend Preprocessing Implementation

* **The fixed sequence of steps implemented:**

Resizing: Standardizing all inputs to 256x256 pixels.

Grayscale Conversion: For intensity-based analysis.

Denoising: Removing artifacts while preserving edges.

Skull Stripping: Removing non-brain tissue (skull/scalp).

Histogram Equalization: Enhancing contrast for clearer tissue boundaries.

Slide 9: [HEADING: PREPROCESSING VISUALIZATION]

* *Add Screenshot of the Preprocessed Output screen here.*

* **Results:**

* The UI displays side-by-side comparisons of original vs. processed images.

* Metadata regarding applied steps is returned via API for user verification.

Slide 10: - Software Testing Methodology

Method: **Black-Box Testing**.

Reasoning: To validate functional behavior (inputs/outputs) from the user’s perspective without analyzing internal code.

Environment: Standard desktop system using a web browser to simulate real-world usage.

Focus: Correctness of image uploads, execution of preprocessing, and UI stability.

Slide 11: Key Test Cases (Authentication & Upload)

| Test ID | Objective | Input | Status |

| --- | --- | --- | --- |

| **TC-01** | Verify Sign-in | Valid Credentials |
Pass

|

| **TC-02** | Invalid Login Handling | Wrong Password |
Pass

|

| **TC-03** | MRI Image Upload | .png / .jpg file |
Pass

|

| **TC-04** | Invalid File Format | .pdf file |
Pass

|

Slide 12: Chapter 5 - Key Test Cases (Preprocessing & UI)

| Test ID | Objective | Expected Result | Status |

| **TC-05** | Preprocessing Execution | Resized, denoised, and skull-stripped image | **Pass**

| **TC-06** | Output Visualization | Display processed data on UI without distortion | **Pass**

Conclusion: The initial 30% of the project (UI and Preprocessing) is stable and has passed all functional tests.

Would you like me to generate specific speaker notes for these slides to help you with your presentation?